

AKSHAY REDDY CHETHIREDDY

Massachusetts, United States | +1 (978) 259-3996 | akshayreddychethireddy15@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Full-Stack Software Engineer with strong foundations in computer science and experience building and deploying production-grade web applications and AI-powered developer tools. Skilled in React, FastAPI, Spring Boot, REST APIs, and PostgreSQL with hands-on experience integrating LLM systems and scalable backend architectures. Passionate about building performant, user-focused software and solving real-world engineering problems.

EDUCATION

University of Massachusetts Lowell

Bachelor of Science, Computer Science (GPA: 3.88/4.0)

May 2025 - May 2026

Lowell, MA

- **Achievements:** Chancellor's List Student
- **Coursework:** Data Structures, Algorithms, Database Systems, Operating Systems, Computer Networks, Software Engineering

TECHNICAL SKILLS

- **Programming Languages:** Python, Java, JavaScript, C++, C
- **Frontend:** React, HTML, CSS, JavaScript, Vite, Axios
- **Backend:** Spring Boot, Restful API, JWT Authentication, Sql, SQLAlchemy, FastAPI
- **Databases:** MySQL, MongoDB, PostgreSQL, SQL
- **Tools & Platforms:** Git/GitHub, Linux, VS Code, Jenkins, Vercel, Render
- **Core Concepts:** Object-Oriented Programming, Data Structures, Algorithms, Problem Solving, Debugging, Testing, Api design
- **AI / Machine Learning:** LangChain, Retrieval Augmented Generation (RAG), LLM Integration, Vector Databases

Work Experience

Indyapay | Software Developer (Internship Program)

Dec 2023 - Dec 2024

- Developed and maintained Java-based web applications with the Spring Framework that processed digital payments and transactions, resulting in reliable service for end-users
- Designed and implemented RESTful APIs with Spring Boot and JWT for user authentication, payment workflows, and third-party integrations, enabling secure and seamless transactions
- Built responsive frontend components with React and CSS, improving page load speed and accessibility for users across devices
- Managed and optimized MySQL and MongoDB databases by adding indexes and refactoring queries, ensuring data integrity and reducing query latency
- Participated in Agile/Scrum ceremonies, sprint planning, code reviews, and CI pipelines using JIRA, Git, and Jenkins, fostering team collaboration and timely delivery
- Debugged production issues, performed root cause analysis, and improved system stability.
- Used Git for version control and collaborated with cross-functional teams to deliver features on time.

Projects

GitHub Repository Analyzer: AI Codebase Intelligence Tool

Dec 2025 - Mar 2026

<https://githubrepoanalyzer.vercel.app>

- Built an AI-powered system that analyzes GitHub repositories to generate architecture summaries, detect technologies, and extract dependencies.
- Implemented a **Retrieval-Augmented Generation pipeline using LangChain, Jina embeddings, and ChromaDB** for codebase search.
- Designed asynchronous repository processing using **Celery and Redis** for scalable background analysis tasks.
- Integrated **Groq LLaMA models** to provide repository insights and interactive codebase Q&A.
- Developed a modern developer interface using **React, TypeScript, and TailwindCSS with streaming chat responses**.
- Containerized services using **Docker** for reproducible local development and deployment.

SkillSync: Learning Momentum Intelligence Platform | <https://skillsync-ar.vercel.app/>

Feb 2026 - Mar 2026

- Built a full-stack SaaS platform for tracking skill progression, learning consistency, and analytics-driven study insights.
- Developed secure REST APIs using **FastAPI, JWT authentication, and SQLAlchemy ORM** for user, skill, and progress management.
- Designed a **PostgreSQL schema and analytics engine** supporting streak tracking, momentum scoring, and activity heatmaps.
- Implemented a **React + TypeScript frontend with Zustand state management and Axios API integration**.
- Deployed the production system using **Render, Vercel, and Neon PostgreSQL**, resolving cross-origin and environment configuration challenges.

Personal Developer Portfolio Website | <https://akshayreddych.vercel.app/>

Dec 2025 - Feb 2026

- Built a personal developer portfolio using **React and modern frontend tooling** to showcase projects and technical skills.
- Designed responsive UI components optimized for both desktop and mobile devices.
- Deployed using **Vercel with CI/CD integration**, enabling rapid updates and high availability.
- Integrated GitHub repositories and live project demos for recruiter visibility.

Languages

- English (Professional)